Guide for using the Basler Camera

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Purpose of document:

- SDK
- Python

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Introduction

Basler cameras are designed for numerous applications that involves machine vision like robotics (which you are taking in this class), automation, industrial inspection and more.

Basler provides several software tools for operating and programming their cameras.

Component	Purpose
Pylon Software Suite	Installer containing the camera drivers, config utils, View, and the development SDK
Pylon Viewer	GUI, graphical application for detecting camera, configs, and viewing camera stream
Pylon SDK	APIs for developing camera applications in C++
pypylon	Official Python interface for controlling Basler cameras
OpenCV	Image processing and display in Python applications

Required hardware and software

1. **Hardware Requirements:**
    a. Basler GigE Vision Camera
    b. Ethernet Cable
    c. PoE

d. Computer running any OS of choice (Windows or Linux)
e. Camera Lens

2. **Software Requirements:**
    a. Basler python Software Suite
    b. Python 3
    c. Pypylon
    d. Opencv-python
    e. numpy

3. **Network Parameters:**
    a. CDA Camera IP - 10.1.1.57
    b. MOSCOW Camera IP - 10.2.1.57

Download links and documentation

Resource	**Link**
Pylon Software Suite	https://www.baslerweb.com/en/software/pylon-software-suite/
Linux installation guide	https://docs.baslerweb.com/software-installation-(linux)
Pylon SDK API docs	https://docs.baslerweb.com/pylonapi/
Pypylon Python package	https://pypi.org/project/pypylon/
Pypylon GitHub Repo	https://github.com/basler/pypylon

Usage

SDK

- Launch the pylonView application

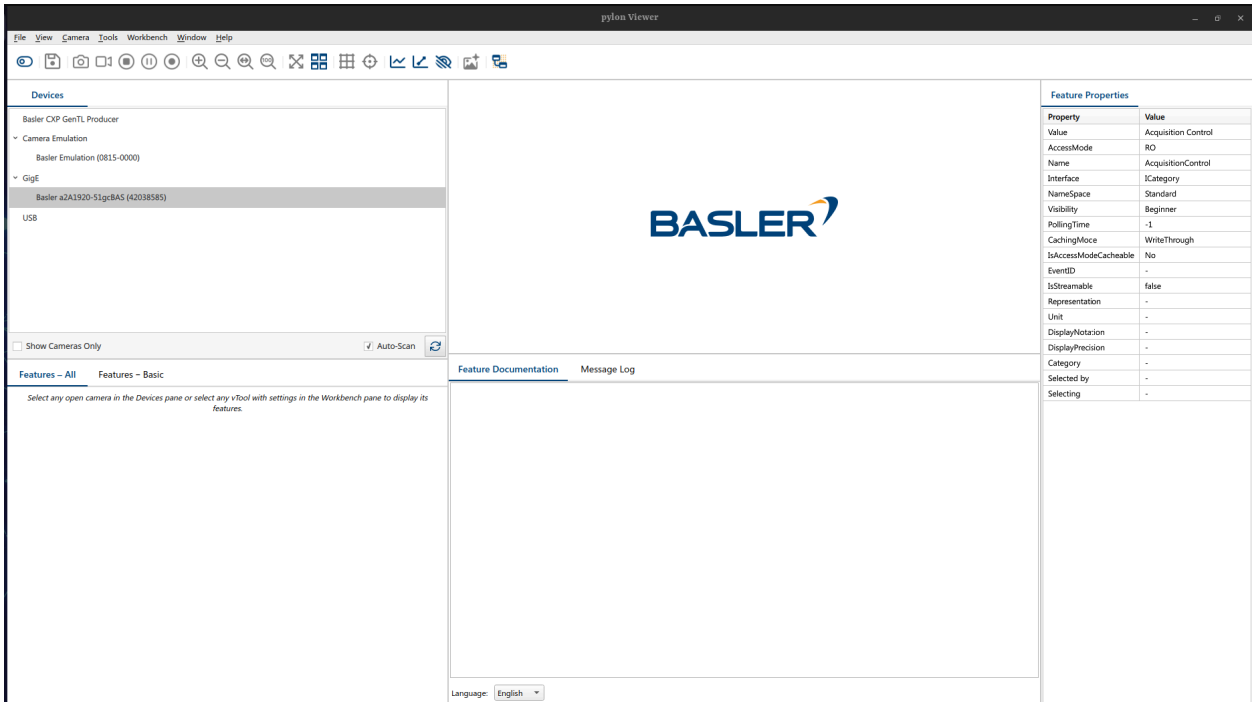


Figure 1:Pylon Viewer - Basler Camera GUI

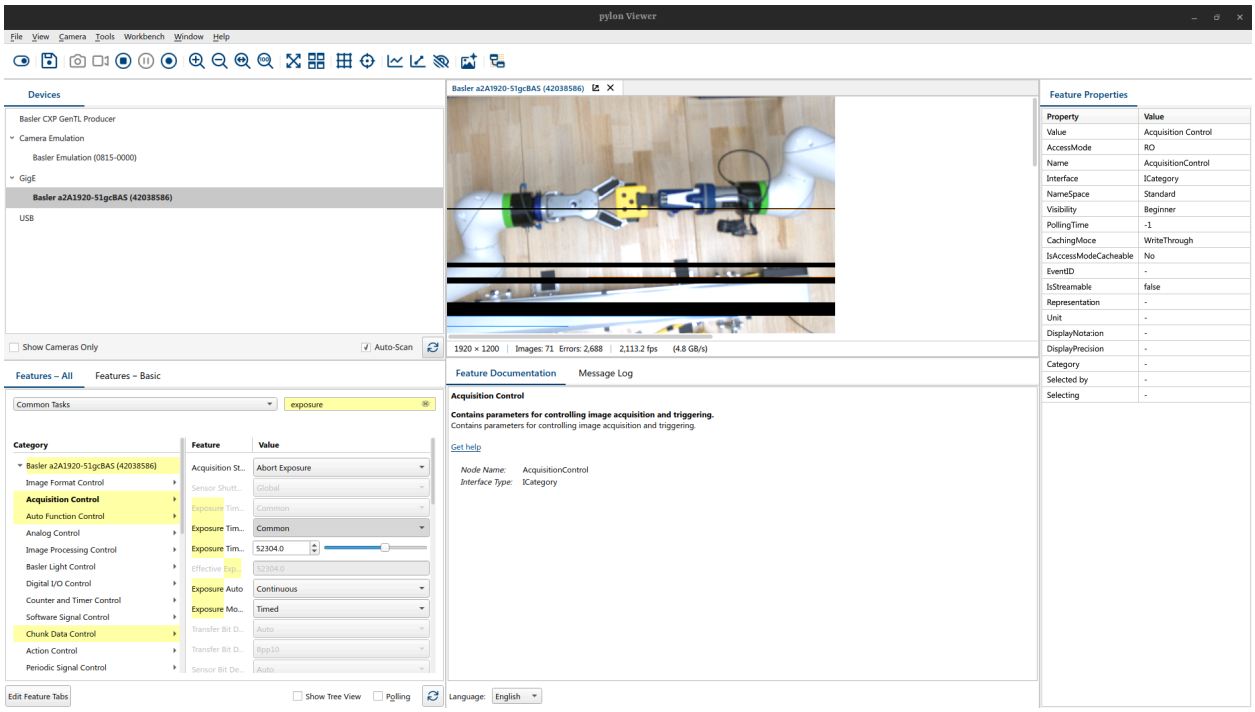


Figure 2: Pylon Viewer with real-time stream

- Under GigE, click on the listed Basler Camera there, in this case, Basler a2A1920-51gcBAS(42038585)

- If the Camera does not show under GigE, then on the nav bar (File, View, Camera)
    - click on Camera, a dropdown will appear
    - click on Add Remote GigE Camera
    - Then enter the IP Address of the Camera based on your campus
    - That will definitely make the camera appear under GigE, if it does not, kindly send me an email.
- After the camera appears, click on the toggle button just under the File (that is the icon just before the save icon). The alternative is to right click on the Basler camera under GigE, and click Open Device
- You can either do a single shot or a continuous shot.
    - Single shot - click on the camera icon next to save, or right click on the Basler camera again and clik on Single Shot
    - Continuous shot - click on the camera icon next to save, or right click on the Basler camera again and clik on Continuous Shot
- If the output is black and you do not see anything in the stream, it is from the Exposure
    - Click on Acquisition Control,
        - Set the Exposure Time Selection to Common
        - Set Exposure Time to 717382.0
        - I noticed those configurations work
        - If you have better configurations, kindly let us know

Python

Below is the sample code for the camera stream

```
# Basler Camera Demo for Robotics - CDA and MOSCOW

import cv2
from pypylon import pylon

CAMERA_IP_CDA = "10.1.1.57"
#CAMERA_IP_MOSCOW = "10.2.1.57"

def main():
   CAMERA_IP = CAMERA_IP_CDA

   try:
       info = pylon.DeviceInfo()
       info.SetPropertyValue('IpAddress', CAMERA_IP)

       #print("info", info)

       tl_factory = pylon.TlFactory.GetInstance()
       camera = pylon.InstantCamera(tl_factory.CreateFirstDevice(info))
```

```
        camera.Open()
        #print(f"Connected to camera at {CAMERA_IP}")

        # modififcation of exposure
        if hasattr(camera, 'ExposureAuto'):
            camera.ExposureAuto.SetValue('Off')

        # exposure is in microseconds
        # target_exposure = 10000
        # the below value seemed to be the sweet spot for the camera
exposure, please change if you find a better
        target_exposure = 717382

        if hasattr(camera, 'ExposureTime'):
            target_exposure = max(camera.ExposureTime.Min,
min(target_exposure, camera.ExposureTime.Max))
            camera.ExposureTime.SetValue(target_exposure)
            print(f"Exposure set to: {camera.ExposureTime.GetValue()} µs")
        # for older GigE models
        elif hasattr(camera, 'ExposureTimeAbs'):
            target_exposure = max(camera.ExposureTimeAbs.Min,
min(target_exposure, camera.ExposureTimeAbs.Max))
            camera.ExposureTimeAbs.SetValue(target_exposure)
            print(f"Exposure set to: {camera.ExposureTimeAbs.GetValue()}
µs")

    except Exception as e:
        print(f"Error: {e}")
        return

    # live stream start
    camera.StartGrabbing(pylon.GrabStrategy_LatestImageOnly)
    converter = pylon.ImageFormatConverter()
    converter.OutputPixelFormat = pylon.PixelType_BGR8packed

    print("Streaming... Press 'q' to quit.")
    while camera.IsGrabbing():
        grabResult = camera.RetrieveResult(5000,
pylon.TimeoutHandling_ThrowException)
        if grabResult.GrabSucceeded():
```

```
            image = converter.Convert(grabResult)
            img = image.GetArray()
            cv2.imshow('GigE cam stream for CDA and MOS', img)
            if cv2.waitKey(1) & 0xFF == ord('q'):
                break
        grabResult.Release()

    camera.StopGrabbing()
    camera.Close()
    cv2.destroyAllWindows()

if __name__ == "__main__":
    main()
```

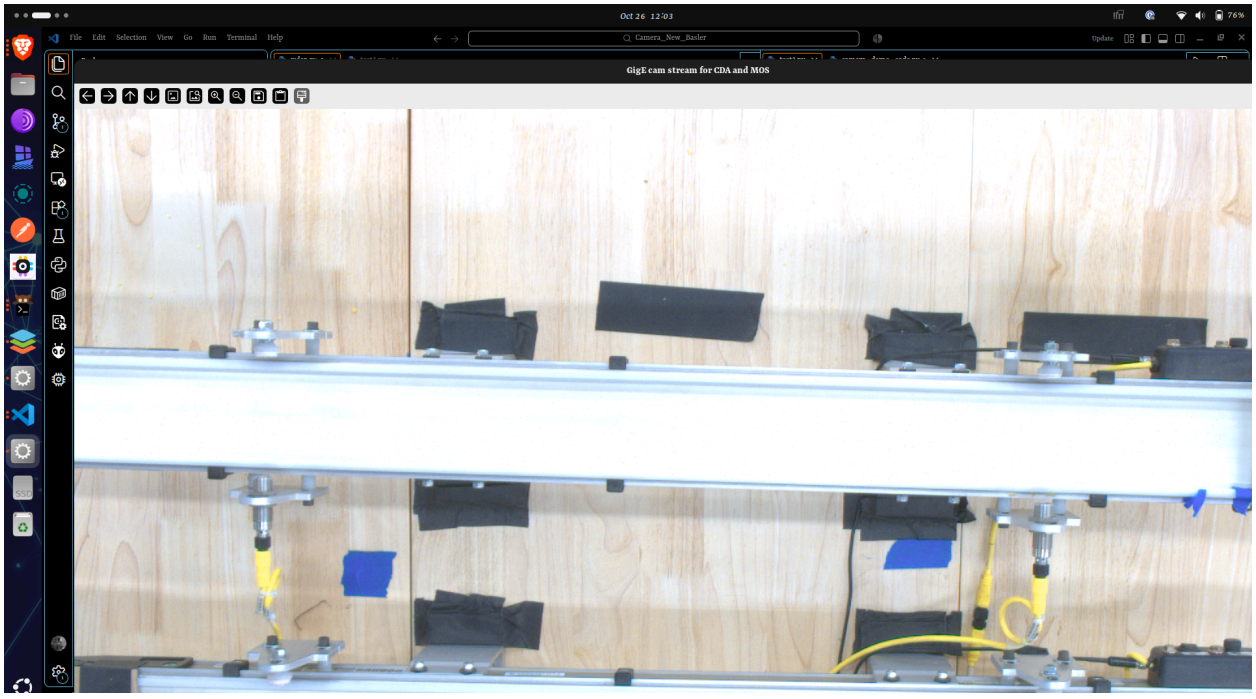


Figure 3: Pypylon python package using OpenCV

Note:

- Single-user access: Only 1 person can use the camera at a time
- Single-application access: Only 1 application can use the camera at a time
- Switching applications: Before using the camera in another application, stop image acquisition and close or disconnect the camera in the current application.

- Connection errors: If the camera is detected but cannot be opened, check whether another application or user is already connected.

Contact

If something does not work as it should, contact

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- Charles Bennington (benn6103@vandals.uidaho.edu)